NINE CAUSES of ERRORS

Causes of software errors

1. **Faulty Requirements Definition**
   - Usually considered the root cause of software errors
   - Incorrect requirement definitions
     - Simply stated, 'wrong' definitions (formulas, etc.)
   - Incomplete definitions
     - Unclear or implied requirements
   - Missing requirements
     - Just flat-out 'missing.' (e.g. Program Element Code)
   - Inclusion of unneeded requirements
     - (many projects have gone amuck for including far too many requirements that will never be used.
     - Impacts budgets, complexity, development time, …

Causes of software errors

2. Client-developer communication failures

- Misunderstanding of instructions in requirements documentation (written / graphical instructions)
- Misunderstanding of written changes during development.
- Misunderstanding of oral changes during development.
- Lack of attention
  - to client messages by developers dealing with requirement changes and
  - to client responses by clients to developer questions

- Very often, these very talented individuals come from different planets, it seems.
  - Clients represent the users; developers represent a different mind set entirely some times!

Causes of software errors

3. Deliberate deviations from software requirements

- Developer reuses previous / similar work to save time.
- Often reused code needs modification which it may contain contain unneeded / unusable extraneous code.

- Book suggests developer(s) may overtly omit functionality due to time / budget pressures.
  - Another BAD choice; System testing will uncover these problems to everyone's dismay!
  - I have never seen this done intentionally – but understand it!
- Developer inserting unapproved 'enhancements' (perfective coding; a slick new sort / search….); may also ignore some seemingly minor features, which sometimes are quite major.
  - Have seen this and it too causes problems and embarrassment during reviews.

Causes of software errors

4. Logical design errors

- Definitions that represent software requirements by means of erroneous algorithms.
  - Yep! Wrong formulas;
  - Wrong Decision Logic Tables;
  - Incorrect descriptions in text
- Process definitions: procedures specified by systems analyst not accurately reflecting the real business process
  - Note: all errors are not necessarily software errors.
  - This seems like a procedural error, and likely not a part of the software system…But they are errors nonetheless!
- Erroneous Definition of Boundary Condition – a common source of errors
  - The "absolutes" like 'no more than' "fewer than," "n times or more;" "the first time," etc.

Causes of software errors

4. Logical design errors

- Omission of required software system states
  - If rank is >= O1 and RPI is numeric, then….easy to miss an action based on the software system state.
- Omission of definitions / procedures concerning reactions to illegal operation of the software system.
  - Including not only code to detect an illegal operation but failure to design the computer software reaction to this: Gracefully terminate, sound alarm, etc. ??

Causes of software errors

5. Coding errors

- Too many to try to list.
  - Syntax errors (grammatical errors)
  - Logic errors (program runs; results wrong)
  - Run-time errors (crash during execution)

- Know the differences. Be also able to supply an example. Examples do not 'define' something, but they may be used to 'support' or 'supplement' a definition.

Causes of software errors

6. Non-compliance with documentation and coding instructions

- Non-compliance with published templates (formats)
- Non-compliance with coding standards
  - (Standards and Integration Branch)
  - Size of program;
    - Other programs must be able to run in environment!
  - Coding: Data Elements and Codes: AFM 300-4;
  - Development Procedures: Required documentation manuals and operating instructions; AFDSDCM 300-8, etc…
- SQA Team: testing not only execution software but coding standards; manuals, messages displayed; resources needed; resources named (file names, program names,…)

Causes of software errors

7. Shortcomings of the Testing Process

- Likely the part of the development process cut short most frequently!
- Incomplete test plans
  - Parts of application not tested or tested thoroughly!
  - Superficial; boundary conditions…
  - Path testing, branch testing … (coverage measures)
- Failure to document and report detected errors and faults
  - So many levels of testing….we will cover.
- Failure to quickly correct detected faults due to unclear indications that there ‘was’ a fault
- Failure to fix the error due to time constraints
  - Many philosophies here depending on severity of error.

Causes of software errors

8. User Interface and Procedure errors

Remember: to the user, the interface is the entire system.

If the Interface is unsatisfactory, this view will be absolutely conveyed 'up the line.'

The 'learnability,' and utility of the interface.

9. Documentation Errors

- Errors in the design documents
  - If Docs do not represent the implemented design, this is trouble for subsequent redesign and reuse
- Errors in the documentation in the User Manuals, Operators Manual, other manuals (Installation…)
- Errors in on-line help, if available.
- Listing of non-existing software functions
  - Planned early but dropped; remain in documentation!
- Many error messages are totally meaningless